Areg Hovumyan

(818) 376-9166 | ahovumyan@cpp.edu | linkedin.com/in/areg-hovumyan | github.com/Arhovumyan | Portfolio | U.S. Citizen

Education

California State Polytechnic University, Pomona — Pomona, CA
Bachelor of Science in Computer Engineering — GPA: 3.6/4.0 — (rising senior) Expected Dec. 2027

Experience and Projects

Electrical/Embedded Systems Member: L'SPACE (Mission Concept) — May 2026 – Present
NASA's student acceleration program - Robotic Lunar Rover Mission — Remote

- Presented the Communication and Data Handling to NASA engineers for a possible mission consideration on the moon.
- Managed to stay within the schedule, cost, and mass limitations (under 250M, 170Kg, no sunlight for 14 days and more).
- Designed the whole rover's Communication and Data Handling(C&DH) architecture by calculating the appropriate battery, figuring out the communication protocols, and choosing the flight software by learning from NASA's previous missions.
- Established (on concept) the communication between the Earth and the Moon, autonomous mission completion in case of redundancy using NeoPan, learned F' to fit the mission under the mission needs.

Software Engineering Team Lead: Autonomous Hexacopter (University Club) — Jun. 2025 – Present
Lockheed Martin sponsored project — Pomona, CA

- Applied pair programming, test-first development, unit-testing, containerized a ROS2 environment with Docker for a team of 8 student-engineers to collaborate, and built CI/CD pipelines to ship features with fewer defects.
- Collaborated with the Electrical team to not overwhelm the drone with object detection and image processing, and worked with the Payload team to drop a water bottle and a beacon mid-flight using servos.
- Presented mapping and object detection improvement, image processing achievement, and correct GPU/CPU utilization to Lockheed Martin engineers during the Preliminary Design Review (PDR).
- Created ROS2 nodes handling the whole mission, including object detection (0.99 mAP@0.5), mapping, 4k image processing (under 0.7s each), and payload release over a target.

Embedded Systems Member: Autonomous Ground Vehicle (University Club) — Feb 2026 - Jun 2026
Northrop Grumman Collaboration Project — Pomona, CA

- Presented telemetry connection between a drone, rover, and a ground station workflow to Northrop Grumman engineers during a Preliminary Design Review (PDR).
- Implemented Feature-driven development and Lean approach to release new code every week and meet all the requirements before any demonstrations or meetings.
- Ran an authorized hardware review of a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and identifying clone silicon.
- Worked alongside with the firmware team to develop bare-metal firmware for a pre-made microcontroller in C, directly manipulating registers (GPIO, timers, interrupts, etc.)

Software Engineer: Autonomous Vehicle Lab (Research) — Apr. 2026 – Present
University Research Lab — Pomona, CA

- Recreating Anduril's "Anvil" interceptor drone, running object detection on a Jetson, establishing autonomous flight through a Pixhawk, and lowering its cost as much as possible.
- Built a closed-loop stepper controller (CL57T) on a microcontroller with CAN telemetry over a TJA1051 transceiver to control a steering wheel through a computer, controller, or a joystick.
- Built CI/CD pipeline in GitHub Actions for the team to catch regressions before merge, and created unit tests using GoogleTest and pytest to improve system reliability.
- Resolved priority-inversion bug by introducing mutex priority inheritance, restoring deterministic timing.
- Utilized AI agents to interface an MPU IMU over I2C on a Nucleo using CubeMX/HAL to stream 9-DOF telemetry.

Technical Skills and Relevant Coursework

Languages and Tools:C, C++, Python, Go, Rust, Java, VHDL, ARM Assembly, Bash, SQL, STM32 (bare-metal, register-level), ESP32, UART, SPI, I2C, DMA, interrupts, timers/PWM, logic analyzers, schematics, ROS2 (Humble/Iron), MAVLink, Gazebo, RPLIDAR, Jetson Orin Nano, YOLOv8/v11, PyTorch, TensorRT, ONNX, OpenCV, Roboflow, computer vision, mathematics, Google Cloud, deep learning, machine learning, compliance, intellectual properly.
Linux, Git, GitHub Actions, Docker, Kubernetes, FastAPI, PostgreSQL, MongoDB, pytest, gtest.
Coursework: Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms, Storage systems, cloud infrastructure, front-end frameworks.
3 Languages: English, Russian, Armenian